S(19): DYADIC REDUCTION AND RELATIVE FORMAL IMMERSIONS

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Here S(d) is the set of primes p for which an elliptic curve over a number field of exact degree d has a point of order p. Equivalently, p divides the order of its torsion subgroup.

Theorem. For elliptic curves over number fields of exact degree 19,
S(19) = {2,3,5,7,11,13,17,19,23,29,31,37,41,43}.
This is a computer-assisted proof. Every required local exclusion and selected-character certificate has passed, and the complete prime and support coverage has no unresolved case.

1. EXISTENCE AND THE DEGREE OF A MODULAR POINT

Every prime through 7 occurs over Q and hence over an extension of degree 19. For p=11,13,17,19, the curve X_1(p) has a rational cusp and genus 1,2,5,7, respectively. Riemann--Roch supplies a rational function with its only pole of exact order 19 at that cusp: both 18 and 19 exceed 2g-2. Hilbert irreducibility then supplies noncuspidal closed points of exact degree 19. The fine moduli interpretation gives the required elliptic curves and points of order p.

The six remaining primes 23,29,31,37,41,43 have explicit witnesses in occurrences.json. verify_occurrences.m proves irreducibility of each polynomial of degree 19, constructs its number field, verifies the nonsingularity of the Tate normal form, and checks that its nonzero point (0,0) is killed by p. Since p is prime, the order is exactly p. The records for 41 and 43 were found in van Hoeij’s low-degree-place data; their validity is checked directly, and completeness of that search is not used.

Conversely, if p>43 occurred over a field K of degree 19, its point on X_1(p) would have degree dividing 19. Mazur’s theorem rules out degree 1, so the point has exact degree 19. Its trace D is an irreducible rational effective divisor of degree 19, with no cusp in its support. If a rational diamond preserves D, its geometric orbit lengths on D are constant, by Galois transitivity, and divide 19.

2. THE IMPROVED DYADIC THEOREM AND THE FINITE PRIME RANGE

We use the previously established dyadic winding-quotient theorem in its integral form, retaining the inclusion 2M subset L subset M for the two cotangent lattices. Equality is not assumed. For m=38=2*19 and M=181, for each u=1,...,180, the 38-by-180 matrix
(epsilon(na)-epsilon(nu/a))_(n,a), 1<=n<=38, 1<=a<=180,
has rank 38 over F_2. Here division is modulo 181 and epsilon is 1 on the residues 91,...,180 and 0 on 1,...,90. Thus the theorem applies when
p > 2*38*181 = 13756.
The complete binary matrices are retained in dyadic_jets.json, and Sage independently reconstructs every entry and verifies all 180 ranks.

For clarity, the factor 2 is used twice in the integral argument. The cotangent image contains 2 times the full coefficient lattice, rather than being assumed saturated. Also, the kernel of reduction on rational dyadic torsion has exponent 2. Doubling the degree 19 divisor gives degree 38 and kills its torsion image. If its elementary symmetric polynomial at the cusp is 1+2B(t), the doubled divisor has polynomial
(1+2B(t))^2 = 1+4(B(t)+B(t)^2).
It lies in the formal neighborhood modulo 4. Target coordinates can be

chosen with linear terms 2x_i and integral terms of order at least 2. For a nonzero x in 4Z_2^38 with minimum valuation k>=2, the linear term has valuation k+1<2k and cannot cancel. Consequently the zero fiber has only the cuspidal point in that neighborhood. The theorem therefore excludes a degree 19 trace whose specialization on X_0(p) is a multiple of one rational cusp, including ramified traces.

Ordinary reduction over F_(2^f) has even group order: its unique nonzero geometric 2-torsion point is rational. In degree 19 its Hasse upper bound is 525737, so an odd prime factor of an ordinary group order is at most 262868. All supersingular elliptic curves in characteristic 2 have j=0. The complete twist orders in residue degree 19 are
 523265=5*229*457, 524289=3*174763, 525313.
They are checked directly by bound_ss.m. The degree 18 twist orders are
 261121,261633,262145,262657,263169,
as checked in the inherited packet. Smaller residue degrees have smaller bounds. A multiplicative identity component has order 2^f-1 or 2^f+1, at most 524289 for f<=19. Additive reduction cannot support these large odd primes. Thus for p>525313 every dyadic specialization lies on a nontrivial multiplicative component and has the same rational cusp image on X_0(p), which the dyadic theorem excludes. Hence p<=525313.

Above 100000, ordinary reduction in degrees<=17 is impossible by parity and Hasse. The degree 17 supersingular orders are
 130561=137*953, 131073=3*43691, 131585=5*26317,
which add no prime above 100000. In degree 18 ordinary reduction requires #E=2p and hence 130561<=p<=131584. In degree 19 ordinary reduction requires #E=2p or 4p; this gives 261421<=p<=262868 or the smaller interval 130711<=p<=131434, already contained in the degree 18 interval. The supersingular lists add 174763 and 525313;262657 is already in the degree 19 ordinary interval. The toric orders add no other prime except 524287. For 524287 the only possible identity-component place has residue degree 19, so it is the unique dyadic place and is unramified. Its trace still specializes to a single rational cusp of X_0(p), and is excluded by the same theorem. Exactly 196 large primes remain for the character screens.

3. GLOBAL DIAMOND CRITERIA AND THE PRIME 163

We use Derickx--Stoll, Prime order torsion on elliptic curves over number fields, Part I: Asymptotics, arXiv:2505.14109v1, Theorem 3.3, Proposition 4.1, Table 2 and Corollary 7.5. The table is used only below 100000, its proved range. The rational 2-primary torsion result in Remark 4.4 is used only within its proved range p<=157.

Write str(p) for the order of the group generated by strange characters. For exact odd degree 19 and p^2>(2^19/325)*19+1, Corollary 7.5 excludes p whenever (p-1)/(2*str(p)) does not divide 19. The two exceptional CM cases in that corollary require even degree. Together with Proposition 4.1 and the bound
 gon_Q(X_1(p)) >= 325*(p^2-1)/65536,
these criteria leave the initial small residual levels
 47,53,59,61,67,71,73,79,83,89,97,101,103,107,109,113,127,131,137,163.
Every inequality and table lookup is evaluated exactly in the ledger.

The last prime has a separate global certificate. Put h=<8> on X_1(163), which has order 27, and B=im(h-1) in J_1(163). The rational homology of B is the sum of the character components of orders 9,27,81. For one primitive character of each order, their coefficient-field degrees are 6,18,54 and their full SIGN-ZERO cuspidal homology dimensions are 24,24,26.

Thus their total rational homology dimension is
 6*24 + 18*24 + 54*26 = 1980 = 2*990.
The odd-prime torsion annihilator A_3 has, after Fricke transport and an invertible diamond factor, matrix T_3-<3>-3. On each of the three components its determinant has odd integral norm. These are exact number-field computations, independently repeated in Sage, including both the determinants and their norms. Therefore A_3 restricted to the actual abelian subvariety B is an isogeny of odd degree. This degree is computed on full rational homology; it is independent of an integral lattice choice and of any isogenous model used to represent the action.

The table proves that (h-1)J_1(163)(Q) is finite. Its points lie in B and are killed by A_3, so this finite group has odd order. The additional 2-primary hypothesis in Proposition 4.1 with ell=2 is therefore satisfied. For a=8 the integer in that proposition is 6, and
 6*19=114 < 325*(163^2-1)/65536.
It follows that hD=D. Noncuspidal h-orbits have length 27, or 9 at the possible j=0 branch points, and neither divides 19. This excludes 163. The certificate is odd_anti163_character_norms_data.m, verified by verify_anti163_norms.py. An additional exact computation on both Betti signs gives odd determinants on the two 990-dimensional anti-invariant spaces. No rational-torsion conjecture at 163 is assumed.

For each of the 196 larger admissible primes put m=(p-1)/2. Choose a prime r with v=v_r(m)>w=v_r(19), and screen every character order n|m for which
 r^(v-w) divides n.
If all these orders are non-strange, then v_r(str(p))<=v-w-1, so v_r(m/str(p))>=w+1 and m/str(p) cannot divide 19. The size inequality in Corollary 7.5 holds throughout this range. There are 1147 required orders:555 inherited completed exact logs and 592 new computations. All required orders have completed certificates, including their independent sequence checks and full divisor-family coverage.

The new calculations use finite-field modular symbols and the cuspidal winding element e’. A scalar sequence u(T_2^i e’), of length 2k for the k-dimensional cuspidal space, has certified minimum recurrence length k. This forces the Krylov span to have dimension k and proves nonvanishing of the winding projection on every relevant characteristic-zero factor. The sparse Hecke action is the native Heilbronn action. The C++ sequence uses exact arithmetic, checks 64 native terms, and asserts the bound
 (k+2)*(q-1)^2 < 2^64
before any accumulation. Sage independently constructs coprime Pade certificates C*S=A mod x^(2k), with C(0)=1, deg(A)<k, deg(C)<=k and max(deg(A)+1,deg(C))=k. Coprimality rules out a shorter recurrence. Random choices search for exact witnesses; they are not a probabilistic premise of the exclusion. The family of every required character order and the unchanged hashes of inherited logs are checked separately.

4. THE COMPLETE DYADIC SUPPORT CENSUS

For the remaining 19 levels, the census enumerates every j-invariant in each residue degree<=19, all twists, and every relevant cyclic p-torsion line. In the rank 2 case it enumerates all p+1 cyclic subgroups; in the cyclic case scalar multiples of one generator suffice. Proper j-fields are retained. Ordinary and supersingular points
are kept separately, with exact closed degrees and complete diamond orbits. Nonrational cusps have degree equal to the order of 2 modulo p and sign. All support-degree multisets of total<=19, including repeated closed points and a rational cusp complement, are enumerated. After the

global exclusion of 163, there are 344 local support types.

The degree 18 proof packet is inherited only after matching actual Tate normal-form diamond orbits in the old and new censuses. Random finite field representatives and their indices are not assumed identical. All 3782 files in its frozen manifest have unchanged hashes. In particular, only the corrected level 97 annihilators are used. There are 14024 inherited ordinary orbits, with the two already known stabilizer exceptions at 113 and 127. The new computation checks 13495 orbits: degree 19 at the old 18 levels and every degree<=19 at 137. Together these exhaust 27519 ordinary orbits in the new local census.

5. GLOBAL SECTION AND PRINCIPAL-DIVISOR CERTIFICATES

Frozen canonical bases are identified by saturated integral q-expansions. The full-lattice assertion uses a geometric bound, not merely a finite rank observation: if an integral truncation of a rational 2-adic differential did not imply integrality, multiplication by a minimal 2-power denominator would give a nonzero mod 2 holomorphic differential vanishing beyond 2g-2. The retained precision rules this out. Fricke identifies these q-form lattices with the differential lattices on the point-level curve and extends integrally because the level is odd.

The verified canonical product dimension 3g-3 also rules out a hyperelliptic special fiber. Its canonical coordinates therefore separate the points used for the orbit keys.

Two modular units identify canonical points and their local jets. Every unit relation is proved globally by its vanishing order exceeding the canonical degree plus the explicitly computed pole divisor degree. At the new level 137 the genus is 715, the q precision is 3500 and the unit pole degrees are 246 and 257. Products of sections are verified beyond the appropriate Sturm bound. Actual elliptic 3-isogenies, with every kernel factor and its trace weight, supply the four Hecke images of a source. The target degrees and weights sum exactly to four times its degree. The stored diamond matrices are globally checked against the actual point action, including the inverse from pullback conventions.

For ordinary sources use alpha*A_3, with A_3=T_3-1-3<3> in the geometric point convention, and
 alpha=1 at 47,59,71;
 alpha=sigma^6-1 at 61;
 alpha=sigma^12-1 at 97;
 alpha=(sigma-1)(sigma^5+1) at 101;
 alpha=sigma-1 otherwise.
Here sigma is the stored primitive diamond generator in the appropriate point convention. The earlier 97 certificates also use the valid alternative (sigma-1)*Phi12(sigma). The stated operators have finite image on rational Jacobian points, and the odd Hecke factor kills that image and all rational cusp contributions.

For the resulting rational divisor U-V over F_2, put L=K^2 at 47,53,61 and L=K at the other levels. Its section dimension exceeds deg(U). Choose f nonzero in H^0(L(-U)). If U-V=div(u), then g=f/u is a nonzero section of L(-V), and for every h in H^0(L(-U)) one has
 h*g in f*H^0(L).
These are necessary linear conditions on g. The exact multiplication tensor and jet evaluations give a matrix for these conditions. Full rank dim(H^0(L)) proves that no such g exists and hence U-V is not

principal. The chosen f and h, section products, jet conditions and full-rank matrices are retained. Every requested multiplicity is asserted to fit the actual retained jet precision.

All 13495 new ordinary sources were checked;13494 give nonprincipal divisors. The one new exception is 137, source 159, degree 17 and closed diamond orbit 4; it is excluded in Section 8. All 7495 new pairs of total 19 pass, including every relative diamond position. At 137 all 369 low pairs of totals 16,17,18 pass as well. Lower-total pairs at the old levels, including the degree 16,17,18 pairs at 131, are inherited.

The new mixed obstructions are 73 with SS6+O13 (21 ordinary orbits),113 with SS7+O12 (9), and 127 with NC7+O12 (4). They use <2>-1, which kills the added supersingular or nonrational cusp divisor. Earlier such obstructions cover the lower degrees. In particular the same inherited O7 and O10 certificates at 73 also cover SS6+SS6+O7 and NC9+O10.

Every one of the 21392 new full-rank obstruction matrices has its rank independently recomputed in Sage:13494 single ordinary matrices,7495 pairs,369 low pairs at 137, and 34 new mixed matrices.

6. CUSPIDAL FORMAL IMMERSIONS AND THE LEVEL 53 REPAIR

Pure cusp disks use the DKSS endomorphism form of Kamienny’s criterion, including its characteristic 2 statement. The maps annihilate the full rational Jacobian groups, with the odd Hecke torsion factor included. Independence is checked in the integral Hecke algebra acting on modular symbols. A relation modulo 2 in the Hecke algebra would give a relation on any stable integral lattice; matrix independence therefore implies the algebra independence required by the criterion. No equality for the cotangent lattice of an optimal winding quotient is assumed. Injective row compression is checked by
reconstructing every original operator row; Sage independently verifies all compressed ranks. Combined-map projections are used only in the valid direction: independence after projection implies independence of the actual operators.

After moving a maximal cusp multiplicity to the first cusp, all other multiplicities are at most 9. Complete binary relation spaces certify the required prefix independence. At 47 a uniform cap 3 block of 69 rows is independent; at most four cusps can have multiplicity>=4. The 31549 covering profiles yield 4208323 nonzero relations, all of prefix cost at least 29>19. An independent exhaustive check also passed all 15754107856 normalized cap 6 patterns. At 59 the uniform cap 4 block has rank 116; at most three cusps can exceed 4. All 3165 profiles pass, with 350115 relations of minimum cost 49. The smaller-index tests at 61,67,71,73 use complete prefix enumeration and separately include all higher multiplicities. The remaining tests use a cap 5 block: at most three cusps can exceed 5, covered by the profiles 19;13+6,12+7,11+8,10+9; and 7+6+6. Every relation in each profile is checked. At 103 and 131 the only allowed degree 19 dependencies are the full simple cusp fibers of degree 17 and 13; their relative treatment is given below.

There is a new exception to the earlier absolute test at 53. On the index 13 quotient X_1(53)/H_2, all high-multiplicity configurations pass, and complete enumeration leaves exactly seven bad normalized cap 6 patterns among 13832534 patterns. The quotient of an exact degree 19 point still has degree 19, since the covering degree is 2. Each rational quotient cusp has the two lifts j and j+13 among the 26 rational cusps

of X_1(53). All 20444 lifts of the seven exceptional patterns are tested on the full curve with the additional maps(<10>-1)A_3 and(<10>-1)A_5. Their joint cotangent rank is 19 at every lift. The full cotangent matrices, including all 19 cusp coefficients, and all 20444 ranks are independently rebuilt in Sage. Thus every lifted zero fiber consists only of its rational cuspidal lift, which excludes a noncuspidal D.

7. THE RELATIVE FORMAL LEMMA AND ITS APPLICATIONS

Let S be smooth of relative dimension d over Z_2, and L a smooth closed formal family of dimension r. Suppose an actual integral morphism Phi:S->A to a smooth group scheme vanishes on L and has cotangent rank d-r at the special-fiber point. Choose normal parameters v for L. Suitable target functions belong to the ideal(v) and have an invertible matrix of linear normal coefficients. Thus they are an invertible matrix of power series times v, and their formal zero locus equals L. The entire zero fiber equals L because Phi vanishes there. This argument is integral and uses no integration denominators at 2.

The maps used here are odd-prime torsion annihilators composed with suitable diamond differences. They annihilate rational Jacobian points and the relevant quotient pullback Jacobians. Fixed rational cusp divisors contribute zero. A fixed local diamond-invariant divisor can also be included, since the remaining base-point term is cuspidal. After Fricke transport their cotangent matrices are
(D_h-1)(T_q-D_q-q),
up to invertible diamond factors. The same actual map and basis act on all support and cusp columns. Coefficient and trace-pairing matrices at a closed point differ by an invertible field-basis transformation.

At 53 the order 13 group acts freely on SS13. The family is R6+pi^*Y and has dimension 1; all 76256 normalized six-cusp complements have normal rank 18. At 61 the order 5 group acts freely on SS15. The family is R4+pi^*Y^(3), of dimension 3, and all 4119 normalized complements have normal rank 16. In both cases R is a nonzero fixed rational cusp divisor, contradicting the support of D.

At 73 take h=<2>, of order 9. The order 3 subgroup fixes the geometric points of SS6 with nontrivial tame tangent action. Their unique local fixed lifts give an h-invariant divisor Z0 of degree 6. The 36 rational cusps form four h-orbits of size 9. Write the cusp complement of degree 13 as R+pi^*E by extracting full fibers. Then deg(E)=0 or 1, and deg(R)=13 or 4. The family Z0+R+pi^*Y^(deg(E)) has the predicted normal rank. For NC9+C10, NC9 itself is an unramified h-fiber, with base rank 8. Extracting an additional full cusp fiber leaves a fixed cusp divisor of degree 10 or 1; the family has dimension 1 or 2. Both cases force a cusp in D.

The complete tangent-kernel computation at 73 uses 148 covering profiles. Modulo the SS or NC base columns, the twenty known relations are the five jet-level full-fiber relations on each of four nine-cusp orbits. Every other relation is represented by an explicitly retained quotient basis. Enumerating all quotient classes and minimizing prefix cost over the known fiber relations gives cost at least 66, exceeding both 13 and 10. The profiles cover one cusp up to 13 or 10, and, for 13, two high cusps 7+6; every other cusp has cap 5. Thus they cover all possible complements. Sage independently proves that the retained bases span the entire relation spaces, not just a sampled set of relations.

At 127 the exceptional O9+NC7+C3 case uses the product of the order 9 and order 7 diamond differences, with the T_2 and T_3 torsion factors. The proved rational cuspidal-torsion result at 127 justifies the T_2 factor. The two quotient pullbacks give a smooth family of dimension 2; the base rank is 14 and all 43680 three-cusp complements have normal rank 17. The diamond orbit lengths 7 and 9 are coprime, so both nonrational supports can be normalized simultaneously. Every relative position is therefore included. The fixed degree 3 cusp divisor gives the contradiction.

At 103 and 131 use the intermediate curves obtained by the odd-order subgroups of orders 3 and 5. Their intrinsic quotient differential spaces are the corresponding invariant subspaces, of genera 120 and 131. The full cusp fibers to X_0 have degrees 17 and 13. All other cusp patterns pass absolute immersion. At a full fiber plus a fixed complement of degree 2 or 6 the intrinsic relative rank is 18, for all 153 or 18564 patterns. The local family is R+pi^*Y and again has R nonzero. Its effective pushforward trace on the intermediate curve would contain a cusp, forcing a cusp in D as well.
All entries of these quotient-cotangent calculations are independently reconstructed in Sage.

8. STABILIZERS AND THE REMAINING RESIDUE DISKS

Let Z=D-C, where C is the rational cusp complement, and define the Jacobian point y=[hZ-Z]. This definition has degree zero even when Z has positive degree. If h has order m and N_h=1+h+...+h^(m-1), then
 m*y = (h-1)[mZ-N_h Z].
The divisor inside the brackets has degree zero. Finite image of (h-1)J(Q) therefore proves that y is torsion, and N_h*y=0 by telescoping. If h fixes the nonrational special-fiber divisor, y reduces to zero; hence 2y=0. If gon_Q(X_1(p))>2*(19+deg(C)), comparison of the effective positive and negative parts of 2(hD-D-hC+C) forces hD-D=hC-C. Their noncuspidal and cuspidal supports are disjoint, so hD=D.

At 73 the actual J_1(73)(Q)[2] is fixed by<2>, as independently audited in the inherited proof. The odd order 9 norm gives N_h*y=9y=y, so y=0. At 89, m=11 is odd and 2y=0, so the displayed identity gives y=my in the actual anti-Jacobian B=im(h-1). The inherited odd-degree annihilator on B kills this rational 2-torsion point, and therefore y=0. These two cases need only gon>19+deg(C).

The exact strict inequalities used are retained in verify_coverage.py. At 73 the largest requirement is gon>19+7=26 for two SS6 divisors; other cases are SS6+NC9 and total degree 18 supersingular/cuspidal supports. The order 9 orbits have lengths 9 or 3, incompatible with 19. At 89, NC11+C8 gives order 11 invariance. At 97, SS16+C3 gives order 4 invariance. At 109, SS9+C10, SS12+C7, doubled SS9 and NC18 give order 18 invariance. At 113, SS7+C12 and total 14 supersingular/cuspidal supports give order 14 invariance. The exceptional ordinary O8, and its repeated case, give order 8 invariance. At 127, pure NC7 and total 14 supports give order 7 invariance, and the exceptional ordinary O9 and its repeated case give order 9 invariance. In all these cases the possible branch orbit lengths still fail to divide 19.

At 113, SS7+O8+C4 uses an involution. Invariance and irreducibility force D to consist of its fixed points, since 19 is odd. These have j=1728 with an i-stable cyclic 113-subgroup. Such a point cannot be real: complex conjugation exchanges the two i-eigenlines. An odd-degree

closed point has a real embedding, which rules out this alternative.
The actual closed-divisor stabilizers of both ordinary O8 orbits and
the SS7 divisor are checked, rather than assuming a pointwise action
from their degrees alone.

At 137, SS17+C2 is fixed by the order 34 diamond<2>. Its possible orbit
lengths 34 or 17 cannot divide 19. The new ordinary exception, source 159
of degree 17 with closed diamond orbit 4, is fixed by<81>=<3>^4 of order 17.
The same strict inequality 2*(19+2)=42<gon(X_1(137)) forces invariance;
its noncuspidal orbits have length 17 and again cannot divide 19.
Both stabilizer assertions are checked on actual Tate normal forms.

9. CERTIFICATES AND COMPLETENESS

The support enumerator includes all 344 local support types. The new
ordinary matrices, every relative pair position, all mixed supports,
and every cusp pattern have either a completed nonprincipal-divisor
certificate, a completed absolute or relative immersion, or an exact
stabilizer argument. The auxiliary 163 argument is global. The complete
prime ledger also includes the dyadic exclusions and all selected
character families. The complete ledger has passed for all 43465 primes in the
finite range and all 344 local support types. It has no unresolved case,
which proves the theorem.

The retained files contain exact inputs, operators, models, points,
jet conditions, rank witnesses and verification scripts. The inherited
packet has a checked 3782-file manifest. All 21392 new obstruction ranks
have independent Sage checks; all cusp compressions reconstruct their
raw operators; the repaired 53 matrices and 20444 ranks are rebuilt
independently; and every relative cotangent map has an intrinsic replay.
REPLAY.txt lists the verification commands, and proof_manifest.json
hashes the required proof inputs.

References used in addition to the previously established dyadic theorem:
Derickx--Kamienny--Stein--Stoll, Torsion points on elliptic curves over
number fields of small degree, Algebra & Number Theory 17 (2023),
Corollary 5.2, Proposition 5.3 and the integral-model discussion.
Derickx--Stoll, Prime order torsion on elliptic curves over number fields,
Part I: Asymptotics, arXiv:2505.14109v1, Theorem 3.3, Proposition 4.1,
Remark 4.4, Proposition 5.2/Table 2 and Corollary 7.5.
The rational-torsion, rational 2-primary reduction-kernel, q-expansion,
Fricke, Eichler--Shimura and Riemann--Roch facts are as specified in the
independently audited inherited proof packet.